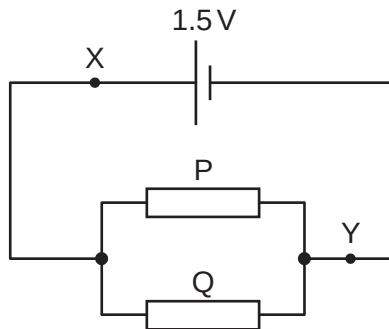


- 22 The diagram shows an electrical circuit consisting of a 1.5 V cell of negligible internal resistance and two resistors P and Q.



Which statement about this circuit is correct?

- A The cell converts 1.5 J of electrical energy to chemical energy for each coulomb of charge passing through it.
- B The energy dissipated per unit charge passing through P and Q is the same.
- C The potential difference across P and the potential difference across Q add up to 1.5 V.
- D The rate of flow of charge at point X is greater than the rate of flow of charge at Y.

Answer: B

Option A is incorrect because the energy conversion inside the cell is from chemical energy

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to electrical energy, and not the other way around.

Option B is correct because P and Q are connected in parallel, thus they have the same potential difference across them. By definition, the work done converting electrical energy to thermal energy per unit charge in each resistor is the same as well.

Option C is incorrect because P and Q are connected in parallel, thus they have the same potential difference across them, which is 1.5 V across each resistor.

Option D is incorrect, because the rate of flow of charge (current) at X is the same as that at Y.

22. In the circuit shown, the cells have negligible internal resistance and the reading on